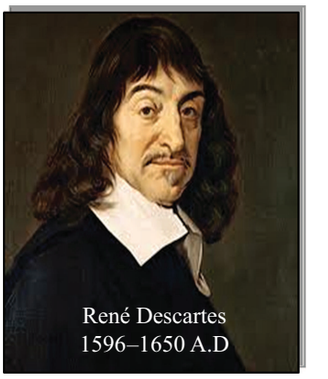
Chapter 6 Two Dimensional Analytical Geometry

"My powers are ordinary. Only my application brings me success ".

Sir Isaac Newton

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6.1 Introduction


René Descartes
1596–1650 A.D

Francois viete(1540-1603) introduced the first systematic *algebraic* notation and contributed to the theory of equations.Two French mathematicians-philosophers **René Descartes** and **Pierre de Fermat** independently founded analytical geometry in the-1630s by adapting Francois viete's algebra to the study of geometric loci. Descartes established analytical geometry as"a way of visualizing algebraic formulas" and developed the coordinate system as "a device to locate points on a plane". His main achievement was to bridge the gap between algebra and geometry. With regard to algebra, he explained in detail that how algebraic equations can be expressed and explained through the use of geometrical shapes. Analytical geometry is a great invention of Descartes and Fermat. Cartesian geometry, the alternative term used for analytical geometry is named after him.

From the17th century onwards, mathematics is being developed in two directions: pure and applied mathematics. One of the first areas of applied mathematic studied in the 17th century was the motion of an object in a straight line. The straight line graphs can be used in the fields of study as diverse as business, economics, social sciences, physics, and medicine. The problem of the shortest line plays a chief and historically important role in the foundations of geometry.

Given a real-world problem, our first task is to formulate the problem using the language of mathematics. Many techniques are used in the construction of mathematical models. Let us see how linear equations (models) can be constructed from a given set of information and solved using appropriate mathematical techniques. Consider some of the real-world, simple problems as illustrated below:

Real life situation 6.1: When a student walks from his house, at an average speed of 6 kmph, reaches his school by ten minutes before school starts. When his average speed is 4 kmph, he reaches his school five minutes late. If he starts to walk to school every day at 8.00 A.M, then how to find (i) the distance between house and the school (ii) the minimum average speed to reach the school on time and time taken to reach the school (iii) the time at which the school starts (iv) the pair of straight lines of his path of walk (Combined equation of two straight lines).

Real life situation 6.2: Suppose the Government has decided to erect a new Electrical Power Transmission Substation to provide better power supply to two villages namely A and B. The substation has